Tramadol

Mechanism of action

- Tramadol is an analgesic. It has multiple actions within the central nervous system, including opiate receptor stimulation and inhibition of the re-uptake of noradrenaline and serotonin.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Moderate to severe pain, particularly if an opiate and/or ketamine is not going to be administered.

Contraindications

- ✕ Known severe allergy.
- ✕ Aged less than 12 years.

Cautions

- ⚠ Tramadol has been taken within the last four hours. Additional tramadol may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold tramadol if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting, the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Aged greater than or equal to 75 years, particularly if there is a previous history of dementia or confusion. Tramadol has anticholinergic activity and this may cause confusion, particularly in the elderly.
- ⚠ Confusion. Tramadol has anticholinergic activity and this may worsen confusion.
- ⚠ Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy and tramadol should usually be withheld.

Tramadol

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 50 mg PO.

Administration

- Administer PO.

Common adverse effects

- Nausea and/or vomiting.
- Feeling light-headed or unusual.
- Sedation.
- Dry mouth.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 4-8 hours.

Usual preparation

- 50 mg capsules.

Pharmacokinetics

- Tramadol is metabolised in the liver and excreted by the kidneys.
- There are no significant effects from liver or kidney impairment on the initial (loading) dose.

Tramadol

Common interactions

- Tramadol has been reported to cause serotonin syndrome in patients taking other medicines or recreational drugs that also raise serotonin levels within the brain. Examples include selective serotonin re-uptake inhibitors (SSRIs), tricyclic antidepressants and Ecstasy. However, this usually only occurs when doses of tramadol higher than 50 mg are taken regularly.

Additional information

- Tramadol is not indicated for pain associated with myocardial ischaemia.
- Do not routinely administer tramadol if an opiate or ketamine is administered because tramadol does not usually provide significant additional pain relief and may worsen side effects. However, tramadol may be administered if the patient has taken tramadol before and found it effective.
- Some patients experience nausea and/or feel unusual with tramadol and may refuse to have it again. These are side effects, not an allergy and are most likely to occur with doses higher than 50 mg administered IV.
- Tramadol has been reported to lower the seizure threshold in patients with epilepsy. However, this usually only occurs with doses higher than 50 mg taken regularly.
- A patient may be administered tramadol (in combination with paracetamol and/or ibuprofen) for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required provided:
  - Adequate pain control is achieved, and
  - The patient is advised to be seen in primary care for a review of their condition.

More Information

Version history

22 November 2022 1.0.1

Update: Aligned with analgesia CPG and caution added.